PFT and Non-Equilibrium Formulas in SCUFF-EM

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1 The SCUFF-EM versions of the BEM system

The usual PMCHW equation reads, in continuous form,

$$\begin{pmatrix} \mathbf{\Gamma}^{\rm EE} & \mathbf{\Gamma}^{\rm EM} \\ \mathbf{\Gamma}^{\rm ME} & \mathbf{\Gamma}^{\rm MM} \end{pmatrix} * \begin{pmatrix} \mathbf{K} \\ \mathbf{N} \end{pmatrix} = -\begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix}_{\parallel}^{\rm inc} \tag{1}$$

or, in discretized form,

$$\underbrace{\begin{pmatrix} \mathbf{M}^{\rm EE} & \mathbf{M}^{\rm EM} \\ \mathbf{M}^{\rm ME} & \mathbf{M}^{\rm MM} \end{pmatrix}}_{\mathbf{M}} \underbrace{\begin{pmatrix} \mathbf{k} \\ \mathbf{n} \end{pmatrix}}_{\mathbf{c}} = -\underbrace{\begin{pmatrix} \mathbf{v}^{\rm E} \\ \mathbf{v}^{\rm H} \end{pmatrix}}_{\mathbf{v}} \tag{2}$$

where e.g.

$$M^{\rm EE}_{mn} = \left\langle \mathbf{b}_m \middle| \mathbf{\Gamma}^{\rm EE} \middle| \mathbf{b}_n \right\rangle.$$

On the other hand, the SCUFF-EM version of (2) reads

$$\underbrace{\begin{pmatrix} \frac{1}{Z_0}\mathbf{M}^{\rm EE} & -\mathbf{M}^{\rm EM} \\ \mathbf{M}^{\rm ME} & -Z_0\mathbf{M}^{\rm MM} \end{pmatrix}}_{\hat{\mathbf{M}}} \underbrace{\begin{pmatrix} \mathbf{k} \\ -\frac{1}{Z_0}\mathbf{n} \end{pmatrix}}_{\hat{\mathbf{c}}} = -\underbrace{\begin{pmatrix} \frac{1}{Z_0}\mathbf{v}^{\rm E} \\ \mathbf{v}^{\rm H} \end{pmatrix}}_{\hat{\mathbf{v}}} \tag{3}$$

The various vectors and matrices in (3) are related to their counterparts in (2) according to

$$\begin{aligned} \hat{\mathbf{M}} &= \begin{pmatrix} \frac{1}{Z_0}\mathbf{M}^{\rm EE} & -\mathbf{M}^{\rm EM} \\ \mathbf{M}^{\rm ME} & -Z_0\mathbf{M}^{\rm MM} \end{pmatrix} \\ &= \underbrace{\begin{pmatrix} \frac{1}{Z_0} & 0 \\ 0 & 1 \end{pmatrix}}_{\equiv\mathbf{A}} \underbrace{\begin{pmatrix} \mathbf{M}^{\rm EE} & \mathbf{M}^{\rm EM} \\ \mathbf{M}^{\rm ME} & \mathbf{M}^{\rm MM} \end{pmatrix}}_{\mathbf{M}} \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & -Z_0 \end{pmatrix}}_{\equiv\mathbf{B}} \end{aligned}$$

and similarly

$$\hat{\mathbf{c}} = \mathbf{B}^{-1}\mathbf{c}, \qquad \hat{\mathbf{v}} = \mathbf{A}\mathbf{v}.$$

Thus (3) is obtained from (2) by **(i)** left-multiplying each side by $\mathbf{A}$, and **(ii)** inserting $\mathbf{B}\mathbf{B}^{-1}$:

$$\underbrace{\mathbf{AMB}}_{\hat{\mathbf{M}}}\underbrace{\mathbf{B}^{-1}\mathbf{c}}_{\hat{\mathbf{c}}} = \underbrace{\mathbf{A}\mathbf{v}}_{\hat{\mathbf{v}}}.$$

Since the correspondence between BEM matrices in the ordinary and SCUFF-EM formulations is

$$\hat{\mathbf{M}} = \mathbf{AMB}$$

the correspondence between *inverse* BEM matrices reads

$$\hat{\mathbf{W}} = \mathbf{B}^{-1}\mathbf{W}\mathbf{A}^{-1}.$$

2 Transformations of compact PFT Formulas

The compact expressions for power, force, and torque in BEM scattering computations generally take the form

$$Q = \mathbf{c}^{\dagger}\mathbf{O}\mathbf{c}$$

where Q is a power, force, or torque, and $\mathbf{O}$ is an overlap matrix.

Using $\mathbf{c} = \mathbf{B}\hat{\mathbf{c}}$, we can express Q in terms of the SCUFF-EM surface-current vector as follows:

$$Q = \hat{\mathbf{c}}^{\dagger}\Big(\mathbf{B}^{\dagger}\mathbf{O}\mathbf{B}\Big)\hat{\mathbf{c}}$$

The matrix that enters this equation is

$$\begin{aligned}\mathbf{B}^{\dagger}\mathbf{O}\mathbf{B} &= \begin{pmatrix} 1 & 0 \\ 0 & -Z_0 \end{pmatrix}\begin{pmatrix} \mathbf{O}^{\mathrm{EE}} & \mathbf{O}^{\mathrm{EM}} \\ \mathbf{O}^{\mathrm{ME}} & \mathbf{O}^{\mathrm{MM}} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -Z_0 \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{O}^{\mathrm{EE}} & -Z_0\mathbf{O}^{\mathrm{EM}} \\ -Z_0\mathbf{O}^{\mathrm{ME}} & Z_0^2\mathbf{O}^{\mathrm{MM}} \end{pmatrix}.\end{aligned}$$